

Assignment – 3

Name: Vaddi Govardhan

Roll No: 142502032

Question - 1:

Identify and fix all the errors in the JSON data from a student management system - 3 Marks

json

```
{
  "students": [
    {
      "id": 101,
      "name": "Sarah Johnson",
      "courses": ["CS101", "MATH200", "ENG150"],
      "gpa": 3.85,
      "active": true,
      "graduation_date": null
    },
    {
      "id": 102
      "name": "Alex Chen",
      "courses": ["CS101" "CS102", "STAT101"],
      "gpa": 3.92,
      "active": True,
      "advisor": undefined,
      "notes": "Excellent student with strong analytical
skills"
    }
  ]
}
```

```
"name": "Maria Rodriguez",
"courses": [],
"gpa": 3.67,
"active": false,
"special_programs": ["honors", "research"],
}
],
"last_updated": "2024-09-15T10:30:00Z"
"total_students": 3
}
```

Answer:

Corrected JSON:

```
{
  "students": [
    {
      "id": 101,
      "name": "Sarah Johnson",
      "courses": [
        "CS101",
        "MATH200",
        "ENG150"
      ],
      "gpa": 3.85,
      "active": true,
      "graduation_date": null
    },
    {
      "id": 102,
      "name": "Alex Chen",
```

```
"courses": [  
  "CS101",  
  "CS102",  
  "STAT101"  
],  
"gpa": 3.92,  
"active": true,  
"advisor": null,  
"notes": "Excellent student with strong analytical skills"  
},  
{  
  "id": 103,  
  "name": "Maria Rodriguez",  
  "courses": [],  
  "gpa": 3.67,  
  "active": false,  
  "special_programs": [  
    "honors",  
    "research"  
  ]  
}  
,  
"last_updated": "2024-09-15T10:30:00Z",  
"total_students": 3  
}
```

Explanation of Errors fixed:

Added missing commas between objects and properties.

Replaced Python/JS literals (True, undefined) with JSON equivalents (true/null).

Fixed missing commas inside arrays and between properties.

Removed trailing commas.

Converted id "103" from string to number for consistency.

Question - 2:

Your web application currently uses this TOML configuration file. - 5 Marks

[server]

host = "0.0.0.0"

port = 8080

debug = false

max_connections = 1000

[database]

url = "postgresql://localhost:5432/myapp"

pool_size = 20

timeout = 30

[logging]

level = "info"

file = "/var/log/myapp.log"

max_size = "100MB"

rotate = true

[[feature_flags]]

name = "new_ui"

enabled = true

rollout_percentage = 25

[[feature_flags]]

name = "analytics"

enabled = false

rollout_percentage = 0

[cache]

```
redis_url = "redis://localhost:6379/0"
```

```
ttl = 3600
```

Analyze the configuration and answer the following questions:

Answers:

a. How many feature flags are currently defined, and which ones are active?

There are 2 feature flags: "new_ui" (enabled = true) and "analytics" (enabled = false).

Active flags: new_ui.

b. What happens when the log file reaches 100MB?

The configuration sets max_size = "100MB" and rotate = true. When the log reaches 100MB it should trigger log rotation (start a new log file and archive/rotate the old one).

c. If you wanted to make the server accessible only from localhost, what should you change?

To make the server accessible only from localhost, change: host = "127.0.0.1" (instead of "0.0.0.0")

d. Calculate the total number of seconds that cached items will remain valid.

Total number of seconds cached items remain valid: ttl = 3600 seconds (given). So cached items remain valid for 3600 seconds (i.e., 1 hour).

e. Explain the difference between the [feature_flags] and [[feature_flags]] syntax.

Difference between [feature_flags] and [[feature_flags]]:

[feature_flags] defines a single table named "feature_flags".

[[feature_flags]] is the inline array-of-tables syntax; each [[feature_flags]] creates and appends a new table to the array 'feature_flags'.

So, the configuration uses [[feature_flags]] to define multiple feature flag entries.